

Problem 4.21

Suppose $A(5) = \$2,500$ and $i = 0.05$.

(a) What is $A(7)$ assuming simple interest?

(b) What is $a(10)$ assuming simple interest?

Solution

Under simple interest,

$$a(t) = 1 + it.$$

Since $i = 0.05$,

$$a(t) = 1 + 0.05t.$$

Also,

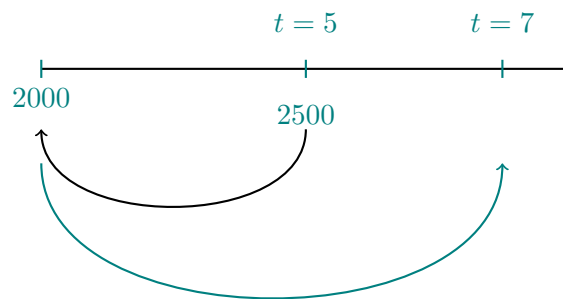
$$A(t) = P a(t) = P(1 + 0.05t).$$

From $A(5) = 2500$ we get

$$2500 = P(1 + 0.05 \cdot 5) = P(1.25).$$

Hence,

$$P = \frac{2500}{1.25} = 2000.$$



For part (a),

$$A(7) = 2000(1 + 0.05 \cdot 7) = 2000(1.35) = 2700.$$

So,

$$\boxed{A(7) = \$2,700}.$$

For part (b), the accumulation function at $t = 10$ is

$$a(10) = 1 + 0.05(10) = 1.5.$$

Thus,

$$a(10) = 1.5.$$